FULL STACK APPLICATION DEVELOPMENT (24SDCS02)

2025-26, Even Semester

Project Problem Statement Submission Form

Sec No- 8	**Team No- 9**	**Project Title- Online Bookstore Management**

1. Abstract

This project focuses on the design and implementation of a RESTful Online Bookstore Management System using Spring Boot, Hibernate ORM, and Maven. The system addresses the need for a centralized backend platform to manage books, categories, and inventory efficiently. Traditional bookstore management often involves manual processes or loosely integrated systems, leading to data inconsistency and poor scalability.
The proposed solution provides CRUD-based REST APIs for managing books and categories, enabling operations such as adding new books, updating stock, deleting records, and retrieving book details with sorting and pagination. Hibernate is used for Object Relational Mapping to ensure seamless interaction between Java objects and the relational database. Spring Data JPA simplifies repository management, while HQL is utilized for advanced queries such as filtering, aggregation, and sorting.
The application follows a layered architecture consisting of Controller, Service, and Repository layers, ensuring separation of concerns and maintainability. Dependency Injection and Inversion of Control are implemented to enhance modularity. Maven handles project build and dependency management, and Git is used for version control.
This system demonstrates real-world backend development practices and provides a scalable foundation for future frontend integration.

2. Module Mapping

Module Name	Assigned Roll Number(s)	Signature
Book Management Module	**2420030032**	
Category Management Module	**2420030265**	
REST API & Controller Layer	**2420030425**	
Database & Hibernate Layer	**2420030433**	

3. Module Details

- **Book Management Module**
  Responsible for CRUD operations on books, including adding, updating, deleting, and fetching books using Spring Data JPA and Hibernate.
- **Category Management Module**
  Handles category creation and mapping of books to categories. Implements HQL queries for sorting and filtering.
- **REST API & Controller Layer**
  Develops REST endpoints using @RestController, HTTP method mappings, @RequestBody, and ResponseEntity to manage responses.
- **Database & Hibernate Layer**
  Creates entity classes, configures generator strategies (AUTO/IDENTITY), and manages ORM operations.

4. Frameworks / Platforms / Software Used

- Java
- Spring Boot
- Spring MVC
- Spring Data JPA
- Hibernate ORM
- Maven
- Git
- MySQL
- Postman (API testing)
- IntelliJ IDEA / Eclipse

Guide Name and Signature (remarks if any) _____